A summary of the goal or purpose of the program in your own words.

The program is designed to fully play a game of Farkle. A player is able to create a meld and then bank that melds score. They are able to reroll dice as long as a valid meld is present. Game will automatically display a farkle if present. User is able to enter there name and will be addressed when it is their turn. Includes a hot hands feature that sees if a player has rolled all six dice as valid numbers and wants to continue their turn. Now added a multiplayer feature and a win screen and condition.

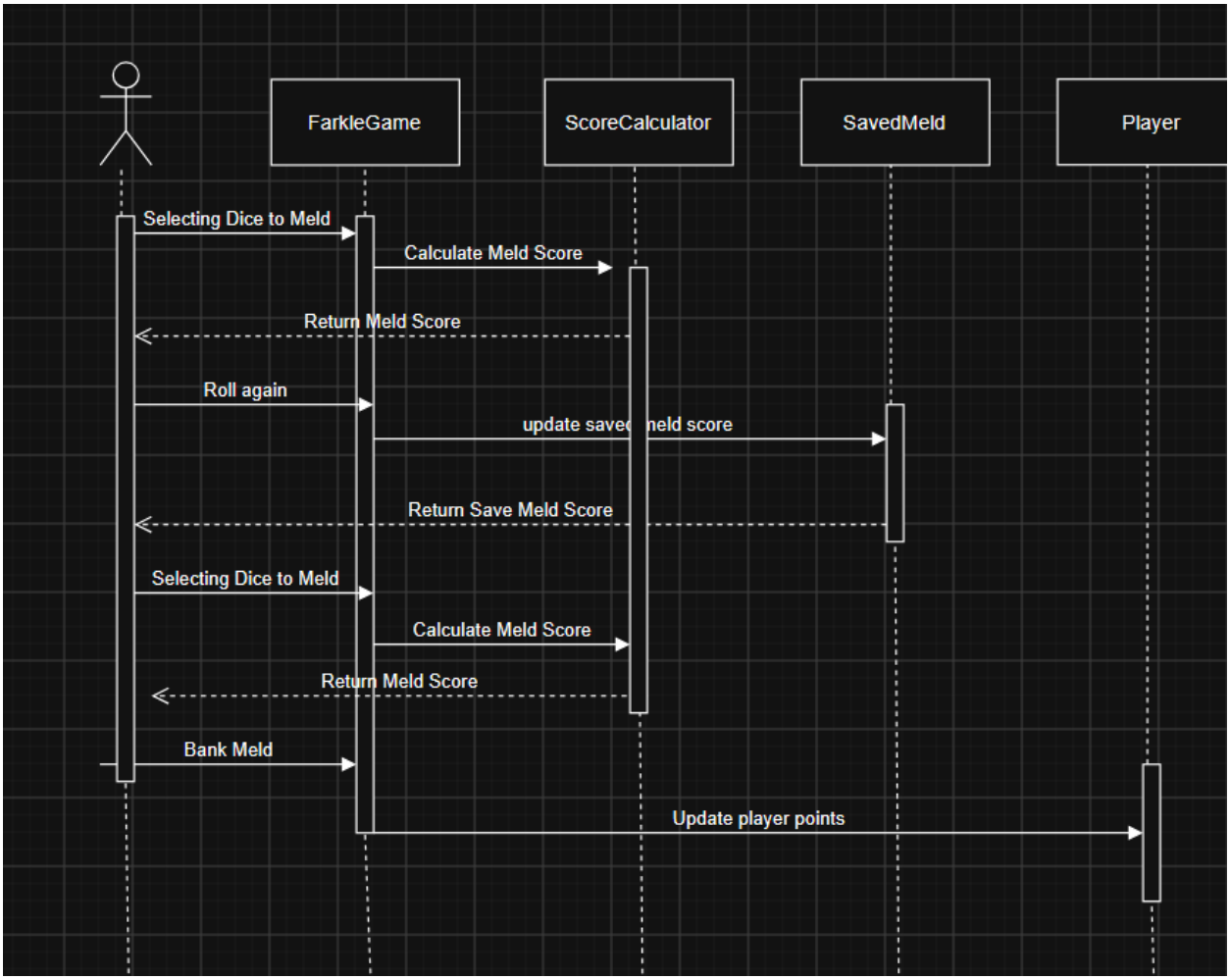
An overview of the general design you chose for your program.

I implemented a valid meld counter to see if they had rolled six valid dice. I split the program into a player, score calculator, dice, farkle game, and saved meld class. The farkle game is what runs the actual turns and uses the other classes for their specific purposes. Score calculator keeps the meld score up to date. Player stores the name and total score a player has accumulated. Dice stores the unmelded die values and creates new rolls. Saved meld stores the saved die that are not being rerolled in the turn.

Unit test Description

Added a new unit test that checks for the full house score condition. It creates a meld that contains a full house and uses the meld calculator to find the score value for that meld. Than it checks to see if the calculated score is equal to the score it should be which in this case is 1500. Returns true if that's correct false otherwise.

Sequence Diagram



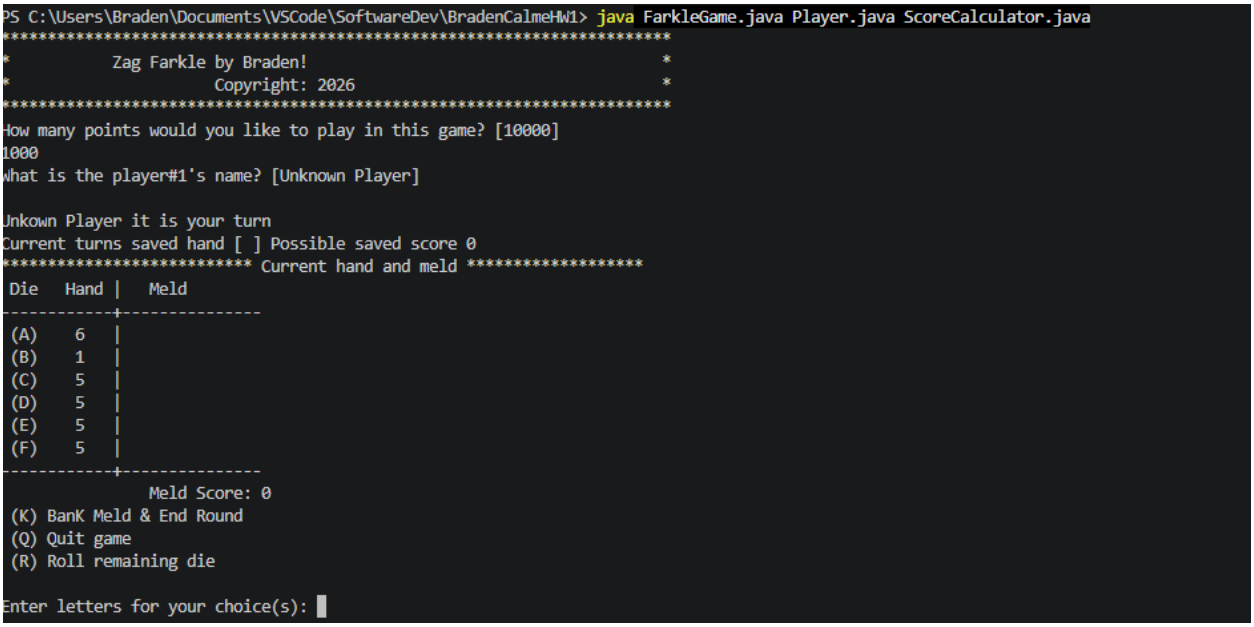
A retrospective of what you would have done differently if you had more time

I would have created my classes differently with more time. My program is extremely jumbled and it is clear I did not know what the final version was going to look like in the beginning. I would simplify each class and have better designed both the methods and variables to meet what I have now learned throughout the semester about classes.

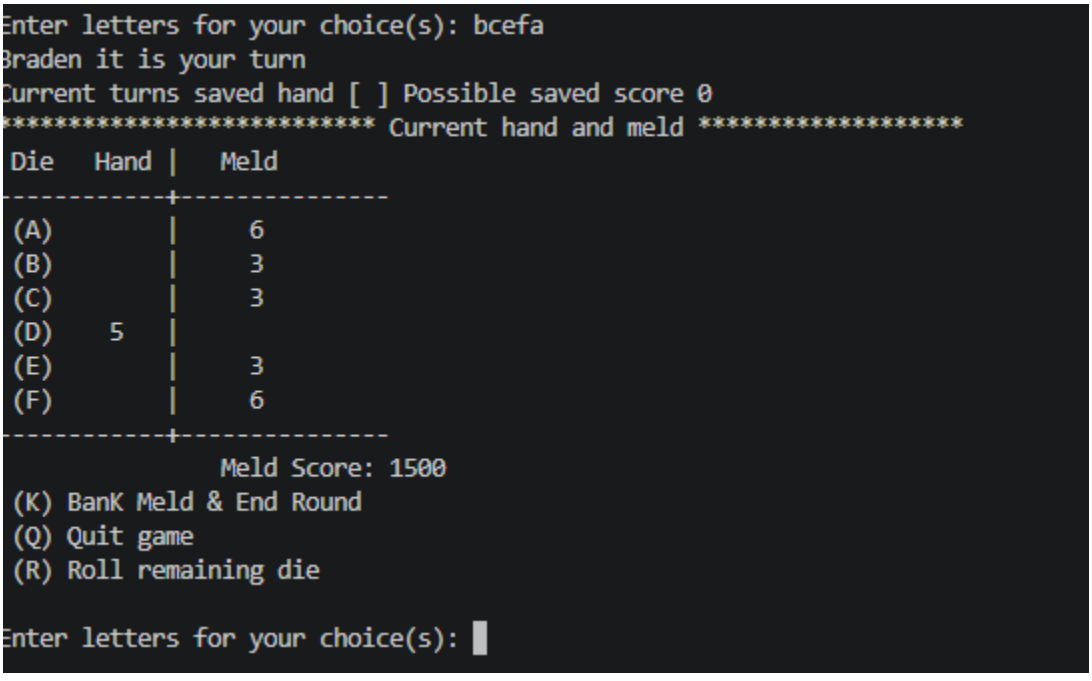
A description of any major design and/or programming issues, why these were the major issues, how you addressed them, and why you addressed them the way you did.

I struggled with creating multiple players. I had originally coded the game to work for a single player. To fix that I created a array of player objects which I than passed the correct player number into the appropriate methods from that array.

Your intro screen with the Requirement #1 game menu



A shot of a Full House combo working



A shot of someone winning the single player game from Requirement #2 (Ended the game once the score was reached)

```
Enter letters for your choice(s): b
Unkown Player it is your turn
Current turns saved hand [ ] Possible saved score 0
************************** Current hand and meld ******************
 Die   Hand |   Meld
------------+--------------
 (A)    6   |
 (B)        |     1
 (C)        |     5
 (D)        |     5
 (E)        |     5
 (F)        |     5
------------+--------------
               Meld Score: 1100
 (K) BanK Meld & End Round
 (Q) Quit game
 (R) Roll remaining die

Enter letters for your choice(s): k

Round over. Total score is now: 1100
```

Shots of entering multiple players, the total scores, and the winner's screen.

```
********************************************************************************
*           Zag Farkle by Braden!                                              *
*                      Copyright: 2026                                         *
********************************************************************************
How many points would you like to play in this game? [10000]
1000
How many players are going to be playing? [1]
2
What is the player#0's name? [Bob]
Braden
What is the player#1's name? [Joe]

Braden it is your turn
Current turns saved hand [ ] Possible saved score 0
************************** Current hand and meld ******************
 Die   Hand |   Meld
------------+--------------
 (A)    6   |
 (B)    3   |
 (C)    3   |
 (D)    5   |
 (E)    3   |
 (F)    6   |
------------+--------------
```

```
-----------+--------------
              Meld Score: 1500
 (K) BanK Meld & End Round
 (Q) Quit game
 (R) Roll remaining die

Enter letters for your choice(s): k
Current total points:
        Bob: 1500
        Joe: 0
Braden has beat the point goal. You have one final chance to beat their score of 1500
Joe it is your turn
Current turns saved hand [ ] Possible saved score 0
************************** Current hand and meld ******************
```

```
Current total points:
        Bob: 1500
        Joe: 1550
Joe has won with a score of: 1550
Braden: 1500
PS C:\Users\Braden\Documents\VSCode\SoftwareDev\BradenCalmeHW1>
```